Assignment

Answer any Five

Sl. No.	Question	Marks
1.	Enlist and discuss the different unit operations that are to be accomplished in a surface mine using discontinuous/cyclic system of mining. Also indicate the equipment to be used for each of the unit operations.	20
2	(a) Derive the effective productivity of a loading shovel. (b) Discuss the factors that influence the effective productivity of a shovel (c) In case of a shovel-dumper operation in a surface mine, prove that the effective productivity of shovel depends on the dumper capacity in case of single spotting condition only, but not in double spotting condition.	5+5+10
3	Estimate the number of drills required in a surface mine for producing 6 million tons of coal per year. Stripping ratio of the mine is 3.2 m^3/te. The coal seam is horizontal. (assume relevant data where ever necessary and mention them)	20
4	(a) Enlist and discuss the geometrical parameters of production benches in surface mines. How are these decided upon? (b) Enlist and discuss the types of failure that may occur in highwall slopes in surface mines.	**(20)**
5	Prepare a statement of heavy earth moving machinery required for a surface coal mine to be worked by shovel-dumper combination to produce 5 million tonnes of coal per year and being worked by shovel dumper combination. The average seam thickness is 10 m and the average stripping ratio is 2.5 $m^3/tonne$. Average haul distance (one way) for both coal and overburden is 2.5 km. (Assume relevant data wherever necessary and mention them)	**(20)**
6	Estimate the HEMM required in a surface Iron ore mine for producing 5.5 million ton of ore per year. The average stripping ratio is 0.8 ton/ton. The mine will be worked by a shovel dumper combination with a bench height of 12 m. (Assume relevant data wherever necessary and mention them)	**(20)**
7	(a) How would you define the effective productivity of a loading shovel? (b) Enlist the factors that affect the effective shovel productivity. (c) In case of shovel – dumper operation in a surface mine show that the	**2+2+8= (12)**

	effective productivity of shovel depends on the dumper capacity in case of single spotting condition only, but not in case of double spotting condition.	
8	How is the bucket capacity of a dragline estimated? Draw a balancing diagram for simple side casting operation of a dragline and derive the expression, in terms of known parameter, for (i) height of spoil heap (ii) reach of the dragline What is meant by MUF of a dragline and what is its physical significance?	**3+7+2= (12)**
9	Estimate the number of drills required to achieve a production of 7.5 million ton/year of coal from a surface mine having coal seam of thickness 10 m. The coal seam is flat and horizontal and the stripping ratio of the mine is $3\,m^3/ton$. (Assume relevant data wherever necessary and mention them)	**(12)**
10	With the help of suitable sketch derive the expressions for (a) optimum spacing between parallel passes and (b) optimum depth of effective loosening in terms of known parameters in case of ripper operation on a horizontal area.	**(12)**
11	(c) What is 'box-cut'? What are its objectives? With the aid of suitable sketches, describe the different types of box cut and also indicate their respective applicability and limitations. (d) Discuss briefly the factors that influence the choice of location of box cut in case of sub-surface deposits.	**(11 + 9)**
12	Enlist and discuss the geometrical parameters of production benches in surface mines. How are these decided upon?	**12**
13	Estimate the number of drills (150 mm dia drill) required in a surface mine producing 5 million te/year of iron ore. Stripping ratio of the mine is 0.2 te/te. (Assume relevant data wherever necessary and mention them)	**(12)**
14	Estimate the number of drills required in a surface mine producing coal at a rate of 5 million te/year. The average seam thickness is 10 m and the average stripping ratio is 3 m^3/te. Dip of the coal seam is 1 in 30. (Assume relevant data wherever necessary and mention them)	**(20)**
15	Enlist and discuss the unit operations that are accomplished in a surface mine using discontinuous/cyclic system of mining. Also indicate the equipment to be used for each of the unit operations.	**(20)**
16	Enlist and discuss the geometrical parameters of production benches in surface mines. How are these decided upon? Enlist and discuss the types of failure that may occur in highwall slopes in surface mines.	**(10+10)**
17	a) Draw a balancing diagram for simple side cast operation of a dragline and derive the expression, in terms of known parameter, for (iii) height of spoil heap (iv) reach of the dragline b) What do you mean by 'minimum tipping load' of a front-end loader? Explain how the concept of minimum tipping load help in selecting suitable bucket sizes in different job conditions.	**(20)**

18	Write short notes and (or) draw labeled diagram wherever necessary for the following		
	(a) Application of high angle conveyor in deep surface mines	(b) Different types of blades used in dozer	
	(c) Schematic presentation of mobile in-pit crushing and conveying system	(d) Concept of tipping load in pay-loaders	
	(e) Types of failure that may occur in highwall slopes in surface mines		